LIBBY — RECOMMENDATIONS

Recommendations — pancreatic-recurrent-kras-g12r-m8f3

Ranked therapeutic options with likelihood, toxicity, mechanism risk, and key references

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Profile snapshot

- **Primary site:** pancreas
- **Histology:** pancreatic adenocarcinoma
- **Stage:** recurrent
- **ECOG:** 1
- **Age band:** 50-59
- **Sex:** unknown
- **Biomarkers:** KRAS G12R mutated; TP53 inactivating mutation; CDKN2A loss; CCND3 alteration; MSI status MSS; TMB 4.1 mut/Mb
- **Efficacy/toxicity weight:** 0.8

11 ranked therapeutic options across 3 targetable features. Biomarker workup steps live in the standalone Target validation paths report.

KRAS G12R-targeting interventions

1. daraxonrasib (RMC-6236) monotherapy

Likelihood of effect: Moderate-to-high in 2L G12 PDAC: ORR 35%, mOS 13.1 mo (n=26, pmid:42090791), but G12R-specific rate unpublished so allele-level confidence is lower.

Toxicity burden: Moderate (rash, diarrhea, stomatitis, fatigue)

Counter-productive MoA:Low (Pan-RAS blockade can select adaptive RTK/MAPK reactivation, but no board persona dissented on mechanism grounds.)

Overall:The only peer-reviewed, indication-matched efficacy on the KRAS axis and the board's unanimous lead; G12R sits inside a pooled G12 estimand rather than its own subgroup.

Key references: PMID 42090791 · PMID 38589574 · PMID 38593348 · NCT05379985 · NCT06625320

2. JYP0015 / ERAS-0015 (pan-RAS molecular glue)

Likelihood of effect: Moderate but unconfirmed: PDAC uORR 40-42% and 14/14 with >=75% ctDNA drop (nct:NCT06895031), from an April 2026 topline release, not peer-reviewed.

Toxicity burden: Not characterized (no DLTs reported; per-term G3+ rates undisclosed)

Counter-productive MoA:Moderate (Pan-RAS glue may drive adaptive RAS-pathway reactivation; ctDNA response has never been shown to translate to RECIST benefit in PDAC.)

Overall:A second pan-RAS shot on the primary axis with a striking but press-release-only ORR; efficacy corroborates the class while the grade-resolved safety table is still missing.

Key references: NCT06895031

3. daraxonrasib + mFOLFIRINOX or gem/nab (RMC-GI-102)

Likelihood of effect: Moderate-to-high on depth of response: chemo-plus-RAS(ON) backbone; RAMP-205 dose-1 5/6 PRs (n=6, abstract; doi:10.1200/JCO.2024.42.16_suppl.4140) previews deeper responses, unproven at scale.

Toxicity burden: Moderate (rash, diarrhea, cytopenia, neuropathy)

Counter-productive MoA:Low (Overlapping chemo/RAS toxicity may force dose compromises that erode the intended combination intensity; no mechanism dissent raised.)

Overall:Combination intensification for a remission-seeking, oxaliplatin-naive patient; a deeper-response ceiling at the cost of stacked, uncharacterized combination toxicity.

Key references: NCT06445062 · PMID 42090791 · doi:10.1200/JCO.2024.42.16_suppl.4140

4. ulixertinib (ERK1/2i) + palbociclib (CDK4/6i)

Likelihood of effect: Low for objective response: best published PDAC readout is stable disease (2 SD, n=26; doi:10.1200/JCO.2021.39.15_suppl.3103); rationale is mechanistic, not efficacy-backed.

Toxicity burden: Moderate (cytopenias, fatigue, rash, nausea)

Counter-productive MoA:Moderate (G12R's lower ERK flux may blunt the ERK-inhibitor arm; overlapping toxicity can force dose reductions that undo the vertical blockade.)

Overall:The only regimen covering KRAS, CDKN2A, and CCND3 together; clean mechanistic logic undercut by a stable-disease-only PDAC signal and G12R-specific uncertainty.

Key references: NCT03454035 · doi:10.1200/JCO.2021.39.15_suppl.3103 · PMID 39437014

5. KST-6051 / FALCON (pan-KRAS ON/OFF inhibitor)

Likelihood of effect: Unknown: no clinical data (first patient dosed April 2026, nct:NCT07458347); class-extrapolation only for the ON/OFF pan-KRAS mechanism.

Toxicity burden: Not characterized (pre-data; no AE table exists)

Counter-productive MoA:Low (Pan-KRAS inhibition risks adaptive pathway reactivation, but no data and no mechanism-grounded dissent were raised.)

Overall:A pre-data pan-KRAS ON/OFF option as mechanistic optionality on the primary axis; G12R-eligible on the KRAS gate, but no efficacy or safety readout exists yet.

Key references: NCT07458347 · PMID 31649109

6. PF-07934040 (pan-KRAS inhibitor, G12R named in eligibility)

Likelihood of effect: Unknown: dose-finding with no responses reported (nct:NCT06447662); ranked for explicit G12R eligibility, not demonstrated activity.

Toxicity burden: Not characterized (dose-finding; only protocol exclusions published)

Counter-productive MoA:Low (Pan-KRAS inhibition risks adaptive reactivation; no data and no mechanism dissent.)

Overall:A first-in-human pan-KRAS program that names G12R in its eligibility criteria; a clean axis backup with no efficacy readout yet.

Key references: NCT06447662

7. JAB-23E73 (pan-KRAS ON/OFF inhibitor)

Likelihood of effect: Unknown: only a qualitative 'early antitumor activity' mention, no response rate (nct:NCT06959615); class-extrapolation on the primary axis.

Toxicity burden: Not characterized (pre-data; no AE table exists)

Counter-productive MoA:Low (Pan-KRAS inhibition risks adaptive reactivation; no data and no mechanism dissent.)

Overall:A second pan-KRAS ON/OFF option as axis redundancy; G12R named preclinically, but the trial is an all-comer basket with only a qualitative activity mention.

Key references: NCT06959615

8. ELI-002 (amphiphile mKRAS peptide vaccine, G12R-encoded)

Likelihood of effect: Low in this setting: strong immunogenicity (84% T-cell response, pmid:40790272) but only in MRD-positive resected disease; no RECIST translation in overt recurrence.

Toxicity burden: Low (injection-site reactions; no G3+ vaccine-attributed events)

Counter-productive MoA:Moderate (In an MSS immune-cold, bulky recurrent tumor, T-cell priming may not overcome the immunosuppressive stroma; setting mismatch is the core risk.)

Overall:The cleanest allele-matched mechanism and safety on the board, foreclosed by an MRD/adjuvant enrollment window that excludes this overt-recurrent patient.

Key references: PMID 40790272 · PMID 38195752 · NCT05726864 · NCT06411691

Pipeline context — not currently enrollable

Other agents in this target class drawn from the case dossier. The patient cannot currently enroll in these trials, but they are included for context on the broader modality landscape.

Intervention	Modality	Phase	Enrolling indication	Recruitment	Trial

zoldonrasib (RMC-7977) + ivonescimab; daraxonrasib + ivonescimab combinations (basket/biomarker)	small_molecule	1/2	Advanced / metastatic solid tumors with RAS mutation (NSCLC, CRC; PDAC eligibility per protocol)	recruiting	NCT07397338
ABO2102 KRAS neoantigen mRNA vaccine ± toripalimab (basket/biomarker)	vaccine	early phase 1	Advanced KRAS-mutated solid tumors — PDAC, NSCLC and others	recruiting	NCT06577532
ASP3082 (setidegrasib, KRAS G12D-selective degrader) + mFOLFIRINOX or NALIRIFOX	small_molecule	3	First-line KRAS G12D-mutated metastatic PDAC	recruiting	NCT07409272
INCB161734 (KRAS G12D-selective) + chemotherapy	small_molecule	3	Previously untreated KRAS G12D-mutated metastatic PDAC (DAWN-303)	recruiting	NCT07522073
Therapy from a menu including abemaciclib (CDK4/6), KRAS-directed agents, PARPi where indicated (off-label)	small_molecule	2	Advanced PDAC selected by patient-derived organoid (PDO) drug sensitivity	not yet recruiting	NCT06813079
avutometinib (VS-6766, RAF/MEK clamp) + defactinib (FAKi) + gemcitabine + nab-paclitaxel (off-label)	small_molecule	1/2	Previously untreated metastatic PDAC with KRAS activating mutation	active not recruiting	NCT05669482
ASP5834 pan-KRAS inhibitor (intravenous)	small_molecule	1	Locally advanced / metastatic solid tumors with KRAS mutations or amplification — NSCLC, PDAC, CRC	recruiting	NCT07094204
MEK inhibitor-based combinations (MEKi + HCQ, MEKi + EGFRi, etc.)	other	n/a	Advanced KRAS G12R altered PDAC	recruiting	NCT05630989
MRTX1133 (KRAS G12D-selective) (discontinued)	small_molecule	1/2	Advanced solid tumors with KRAS G12D mutation	terminated	NCT05737706

Evidence in detail

daraxonrasib (RMC-6236) monotherapy

Wolpin et al. *New England Journal of Medicine* 2026

- **Disease context:Previously treated advanced RAS-mutated pancreatic ductal adenocarcinoma (PDAC)** (2L+) — RMC-6236-001 (NCT05379985) phase 1/2 PDAC cohort; 168 PDAC patients dosed at 300 mg or lower; second-line RAS G12 subgroup analyzed at the 300-mg dose (n=26). G12X variants enrolled include G12D, G12V, G12R, and Q61H — the patient's G12R is in scope.
- **n:** 168

- **Toxicities:any treatment-related AE** (grade any): 96% (n=168); **any treatment-related AE** (grade >=3): 30% (n=168); **rash** (grade any): 91% (n=127); **rash** (grade >=3): 8% (n=127); **diarrhea** (grade any): 48% (n=127); **diarrhea** (grade >=3): 2% (n=127); **nausea** (grade any): 43% (n=127); **stomatitis** (grade any): 31% (n=127); **stomatitis** (grade >=3): 3% (n=127); **vomiting** (grade any): 31% (n=127); **fatigue** (grade any): 20% (n=127)
- **Efficacy:ORR** 35%; **DoR** 8.2 months; **PFS** 8.5 months; **OS** 13.1 months; **TRAE_rate** 96%; **TRAE_rate** 30%

daraxonrasib + mFOLFIRINOX or gem/nab (RMC-GI-102)

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ulixertinib (ERK1/2i) + palbociclib (CDK4/6i)

O'Hara et al. *Clinical Cancer Research* 2025

- **Disease context:Histology-agnostic advanced solid tumors with CDK4 or CDK6 amplification (NCI-MATCH subprotocol Z1C)** (2L+) — 38 eligible/treated patients with confirmed CDK4 or CDK6 amplification (≥7 copies). 25 patients in the primary analysis cohort. Pancreatic cancer was not the dominant histology; the read-out is for the CDK4/6 amplification call, not CDKN2A loss.
- **n:** 38
- **Toxicities:neutropenia** (grade >=3): (n=38)
- **Efficacy:ORR** 4%; **PFS** 2.0 months; **OS** 8.8 months

ELI-002 (amphiphile mKRAS peptide vaccine, G12R-encoded)

Wainberg et al. *Nature Medicine* 2025

- **Disease context:Resected PDAC and CRC with MRD positivity post locoregional therapy (AMPLIFY-201 final follow-up)** (adj) — Final report at 19.7 months median follow-up; n=25 (20 PDAC, 5 CRC); KRAS G12D / G12R targeted.
- **n:** 25
- **Toxicities:** Maintained tolerability at extended follow-up; 71% of evaluable patients induced both CD4 and CD8 KRAS-specific T-cell responses.
- **Efficacy:RFS** not reached months; **RFS** 3.02 months; **OS** not reached months; **OS** 15.98 months; **biomarker** 71%

Pant et al. *Nature Medicine* 2024

- **Disease context:Resected PDAC and CRC with ctDNA / tumor-marker MRD positivity after locoregional therapy (AMPLIFY-201)** (adj) — MRD-positive after curative-intent surgery and standard adjuvant therapy; 25

patients (20 PDAC, 5 CRC); KRAS G12D or G12R required. The patient's G12R is in scope, but the patient is overt-recurrent rather than MRD-only.

- **n:** 25
- **Toxicities:dose-limiting toxicity** (grade >=3): 0% (n=25); **injection-site reaction** (grade any): (n=25)
- **Efficacy:biomarker** 84%; **RFS** 16.33 months; **RFS** not reached months; **RFS** 4.01 months; **AE_rate** 0%

Germline BRCA / HRD-targeting interventions

1. olaparib maintenance after platinum induction (POLO regimen) — gated on germline BRCA1/2 + platinum

Likelihood of effect: Contingent: strong PFS in gBRCA disease (HR 0.53, pmid:31157963) but flat interim OS (HR 0.91), and inapplicable unless germline-positive and platinum-induced.

Toxicity burden: Moderate (anemia, fatigue, nausea; MDS/AML long-tail)

Counter-productive MoA:Low (PARP maintenance carries no mechanism that blunts the therapeutic goal; the dissent was preference-fit, not mechanism.)

Overall:The only guideline-listed drug for this patient, but double-gated on an unordered germline result and an absent platinum induction, so it cannot start today.

Key references: PMID 31157963 · PMID 33970687

2. rucaparib maintenance — gated on germline/somatic BRCA1/2 or PALB2 + platinum

Likelihood of effect: Contingent: mPFS 13.1 mo, ORR 41.7% in HRD-positive PDAC (n=46, pmid:33970687), but inapplicable unless germline/somatic-positive and platinum-induced.

Toxicity burden: Moderate (anemia, fatigue, thrombocytopenia; MDS/AML long-tail)

Counter-productive MoA:Low (PARP maintenance carries no goal-blunting mechanism; contingency, not mechanism, limits it.)

Overall:A PARP backup that broadens the biomarker net to somatic BRCA and PALB2; same germline-plus-platinum double-gate as olaparib, on phase 2 evidence.

Key references: PMID 33970687

Pipeline context — not currently enrollable

Other agents in this target class drawn from the case dossier. The patient cannot currently enroll in these trials, but they are included for context on the broader modality landscape.

Intervention	Modality	Phase	Enrolling indication	Recruitment	Trial

olaparib 300 mg PO BID maintenance (off-label)	small_molecule	3	Germline BRCA1/2-mutated metastatic PDAC, maintenance after first-line platinum	completed	NCT02184195
olaparib maintenance (off-label)	small_molecule	2	Resected PDAC with germline or somatic BRCA1/2 or PALB2 mutation	recruiting	NCT04858334

Evidence in detail

olaparib maintenance after platinum induction (POLO regimen) — gated on germline BRCA1/2 + platinum

Golan et al. *New England Journal of Medicine* 2019

- **Disease context:Germline BRCA1/2-mutated metastatic PDAC, maintenance after first-line platinum-based chemotherapy (POLO)** (maintenance) — 154 randomized; gBRCA1/2 pathogenic variant; no progression after ≥16 weeks of first-line platinum. The patient has no germline data yet and was treated with adjuvant FOLFIRI (no platinum exposure), so both biomarker and platinum-exposure gates are open.
- **n:** 154
- **Toxicities:any treatment-related AE** (grade >=3): 40% (n=92); **any treatment-related AE** (grade >=3): 23% (n=62); **discontinuation for AE** (grade any): 5% (n=92)
- **Efficacy:HR_PFS** 0.53 HR (95% CI 0.35-0.82); **PFS** 7.4 months; **PFS** 3.8 months; **HR_OS** 0.91; **OS** 18.9 months; **OS** 18.1 months; **ORR** 23.1%; **ORR** 11.5%

Reiss et al. *Journal of Clinical Oncology* 2021

- **Disease context:Platinum-sensitive advanced PDAC with germline or somatic BRCA1/2 or PALB2 pathogenic variant** (maintenance) — 46 enrolled, 42 evaluable; germline BRCA1 n=7, BRCA2 n=27, PALB2 n=6; somatic BRCA2 n=2. Broader than POLO by including PALB2 and somatic variants.
- **n:** 46
- **Toxicities:anemia** (grade any): 74% (n=46); **anemia** (grade >=3): 22% (n=46); **nausea** (grade any): 48% (n=46); **nausea** (grade >=3): 2% (n=46); **increased ALT** (grade any): 47% (n=46); **increased ALT** (grade >=3): 4% (n=46); **fatigue** (grade any): 45% (n=46); **fatigue** (grade >=3): 4% (n=46); **thrombocytopenia** (grade any): 39% (n=46); **thrombocytopenia** (grade >=3): 4% (n=46); **dysgeusia** (grade any): 37% (n=46)
- **Efficacy:PFS_rate** 59.5%; **PFS** 13.1 months; **OS** 23.5 months; **ORR** 41.7%; **DoR** 17.3 months

rucaparib maintenance — gated on germline/somatic BRCA1/2 or PALB2 + platinum

Reiss et al. *Journal of Clinical Oncology* 2021

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- **n:** 46
- **Toxicities:anemia** (grade any): 74% (n=46); **anemia** (grade >=3): 22% (n=46); **nausea** (grade any): 48% (n=46); **nausea** (grade >=3): 2% (n=46); **increased ALT** (grade any): 47% (n=46); **increased ALT** (grade >=3): 4% (n=46); **fatigue** (grade any): 45% (n=46); **fatigue** (grade >=3): 4% (n=46); **thrombocytopenia** (grade any): 39% (n=46); **thrombocytopenia** (grade >=3): 4% (n=46); **dysgeusia** (grade any): 37% (n=46)
- **Efficacy:PFS_rate** 59.5%; **PFS** 13.1 months; **OS** 23.5 months; **ORR** 41.7%; **DoR** 17.3 months

CDKN2A-loss / MTAP-targeting interventions

1. anvumetostat (AMG 193, MTA-cooperative PRMT5i) — gated on MTAP co-deletion

Likelihood of effect: Contingent and unproven: synthetic-lethal rationale (pmid:26912361, 26912360) is preclinical; no PDAC efficacy, and it collapses if MTAP is retained.

Toxicity burden: Not characterized (no clinical safety row; PRMT5-class myelosuppression/GI expected)

Counter-productive MoA:Moderate (Rationale collapses entirely if MTAP is retained rather than co-deleted; combination-arm toxicity uncharacterized.)

Overall:A synthetic-lethal PRMT5 option for the CDKN2A-loss axis, potentially co-targeting KRAS in one trial; entirely gated on an unrun MTAP reflex test.

Key references: NCT06360354 · PMID 26912361 · PMID 26912360 · PMID 37552839

Pipeline context — not currently enrollable

Other agents in this target class drawn from the case dossier. The patient cannot currently enroll in these trials, but they are included for context on the broader modality landscape.

Intervention	Modality	Phase	Enrolling indication	Recruitment	Trial
BMS-986504 (navlimetostat / PRMT5 inhibitor) ± daraxonrasib, immunotherapy, or chemotherapy (basket/biomarker)	small_molecule	2	Solid tumors (GBM, melanoma, NSCLC, PDAC) with homozygous MTAP deletion	not yet recruiting	NCT07492680